SAP On-Premise to Cloud ERP Transition: Enterprise Modernization Case Study

Consulting Case Study | Manufacturing & Industrial Products Industry

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Executive Summary (TLDR)

A global industrial manufacturer running a heavily customized, 15-year-old SAP ECC on-premise environment faced rising maintenance costs, a looming end of mainstream SAP support, and an inability to roll out new business models (subscription services, real-time inventory visibility) on its legacy stack. A structured migration to SAP S/4HANA Cloud — paired with a "clean core" strategy to strip out a decade of custom code — cut month-end close time from 12 days to 3 days, reduced total cost of ownership by 28%, and enabled three new digital revenue streams within 18 months of go-live.

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1. Background

A global industrial manufacturer had run SAP ECC on-premise for over 15 years, with the system deeply customized to accommodate regional finance rules, plant-specific manufacturing workflows, and legacy integrations built up over multiple acquisitions. The environment worked, but it had become a liability: over 40% of core modules contained custom code that no longer matched SAP's standard processes, patches took months to test and deploy safely, and SAP's announced end of mainstream maintenance for ECC created a hard deadline for action.

Beyond the technical debt, the business was constrained. Finance couldn't close the books in under two weeks, supply chain planners lacked real-time visibility across plants, and new business initiatives — like equipment-as-a-service subscription offerings — couldn't be supported without extensive custom development. Competitors running modern cloud ERP platforms were bringing new products to market faster and operating with leaner finance and IT teams. Leadership had to decide between a costly re-hosting of the existing customized system, a full re-implementation on SAP S/4HANA Cloud, or a hybrid path that preserved critical customizations while modernizing the core.

2. Methodology

A 6-phase approach combining ERP transformation and organizational change methodology was applied:

- **Phase 1 – Current-State Assessment:** Inventory custom code, integrations, and business processes across all divisions; quantify which customizations were business-critical versus historical workarounds.
- **Phase 2 – Clean Core Strategy Design:** Define which processes would move to SAP standard functionality and which required extensions built via the supported side-by-side extensibility model rather than core modifications.
- **Phase 3 – Target Architecture & Data Design:** Design the S/4HANA Cloud landscape, master data model, and integration architecture connecting retained legacy and third-party systems.
- **Phase 4 – Process Redesign & Fit-to-Standard Workshops:** Run fit-to-standard workshops per business function to align processes to SAP best practices, minimizing new custom build.
- **Phase 5 – Phased Migration:** Migrate business units and regions in waves, starting with a lower-complexity pilot entity before scaling to global operations.
- **Phase 6 – Stabilize & Optimize:** Retire legacy infrastructure, decommission redundant point solutions, and continuously tune adopted processes post go-live.

3. Key Considerations

- A full "lift and shift" of existing customizations onto the cloud platform was rejected; it would have carried forward the same technical debt and defeated the purpose of moving to a standardized, upgrade-friendly core.
- Roughly 60% of existing custom code was found to replicate functionality SAP already delivered out of the box, once processes were re-examined rather than assumed to be unique requirements.
- Finance, supply chain, and plant operations each had different appetites for standardization, requiring function-by-function negotiation rather than a single blanket policy.

4. Sample Deliverables

- Custom code and process inventory with disposition (retire, retain via extension, replace with standard)
- Clean core architecture blueprint and integration map
- Fit-to-standard workshop outputs and process design documents by function
- Phased migration and cutover plan with entity-by-entity sequencing
- Change impact and training plan for finance, supply chain, and plant users

5. Strategic Recommendations

The recommended path was a clean core migration to SAP S/4HANA Cloud: adopt SAP standard processes wherever they met 80% or more of a function's needs, isolate any remaining unique requirements into supported extensions outside the core, and phase the rollout by business entity to contain risk. This approach traded a small amount of process customization for dramatically lower long-term maintenance cost and preserved the organization's ability to adopt future SAP innovations without re-engineering the core each time.

6. Culture and Change Management

Long-tenured finance and plant staff had built deep expertise around the legacy system's quirks and were skeptical that standard processes could handle their specific requirements. The program addressed this by involving process owners directly in fit-to-standard workshops rather than presenting a finished design, and by demonstrating standard-process outcomes using the organization's own data during the pilot wave, which converted several vocal skeptics into internal champions for later waves.

7. Technology and Cloud ERP Modernization

Rather than replicating customizations in the cloud environment, a clean core approach was used: standard SAP S/4HANA Cloud processes handled the large majority of finance, procurement, and manufacturing workflows, while a small set of genuinely unique requirements were addressed through side-by-side extensions connected via released APIs, keeping the core upgradeable and reducing the ongoing cost of adopting future SAP releases.

8. Operational Efficiency

Standardizing on out-of-the-box processes and automating previously manual reconciliation steps significantly reduced month-end close effort, while real-time data replication across plants eliminated the delays and manual consolidation that had previously slowed supply chain reporting.

9. Regulatory Considerations

Statutory and tax reporting requirements across multiple jurisdictions were validated against SAP S/4HANA Cloud's localized compliance content, with region-specific reporting configured using supported localization packages rather than custom code, reducing the risk of falling out of compliance as regulations changed.

10. Risk Identification

Key risks included data migration errors during cutover, the possibility that a small number of "must-have" customizations were incorrectly deemed replaceable by standard processes, and the risk of user adoption issues if the change management pace outstripped training. Each was mitigated through parallel-run testing before cutover, a formal exception process for challenging standardization decisions, and role-based training delivered close to each wave's go-live date.

11. Cost-Benefit Analysis

The clean core cloud migration carried higher upfront implementation cost than a straight technical upgrade of the existing on-premise system, but eliminated the recurring cost of maintaining custom code against SAP's release cycle. The business case showed payback within roughly two years, driven by lower total cost of ownership, reduced IT maintenance headcount needs, and faster time-to-value on subsequent SAP feature releases.

12. Monitoring and KPIs

Core KPIs tracked through each migration wave included month-end close duration, percentage of processes running on SAP standard functionality, number of active custom code objects, system uptime during cutover, and user adoption rates by function, allowing the team to validate readiness before each subsequent wave.

13. Integration Roadmap

Migration proceeded in three waves — a lower-complexity pilot business unit, a mid-size regional cluster, and finally the largest global divisions — with each wave's KPIs and lessons learned reviewed before the next wave's scope was finalized, allowing the extension architecture and training approach to be refined based on real usage.

14. Return on Investment

Within 18 months of the first wave's go-live, month-end close time dropped from 12 days to 3 days, total cost of ownership fell by 28% as legacy infrastructure and custom-code maintenance were retired, and the standardized core enabled the launch of three new digital revenue streams that would not have been feasible on the prior legacy environment.

15. Establishing Governance

A joint IT-Finance-Business Process steering committee was established to own standardization decisions, arbitrate exception requests for custom extensions, and ensure each wave's scope balanced technical feasibility with business readiness.

16. Implementation Playbook

A standardized wave-migration playbook was created covering data migration checklists, fit-to-standard workshop templates, cutover and parallel-run procedures, and role-based training sequencing, reducing variability in how each business unit's migration was executed.

17. Additional Considerations

The clean core architecture was explicitly designed to make future SAP innovations — new modules, AI-driven features, and regulatory updates — adoptable with minimal rework, avoiding the trap of rebuilding technical debt in the new cloud environment that had made the original on-premise system difficult to maintain.

18. Key Findings and FAQs

Why not just migrate the existing customizations as-is to the cloud? Carrying forward the same customizations would have replicated the maintenance burden and upgrade friction that made the on-premise system a liability in the first place, undermining the core benefit of moving to a standardized cloud platform.

What was the single biggest enabler of the cost reduction? Reducing the volume of custom code by re-validating which "unique" requirements could actually be met by SAP standard processes, which lowered both migration effort and ongoing maintenance cost.

Is a clean core approach reusable for other organizations migrating from SAP ECC? Yes — the fit-to-standard and side-by-side extensibility pattern is broadly applicable for any organization facing SAP's ECC end-of-maintenance deadline, particularly those with significant legacy customization.